

TIC 237115819

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.792059 d
Epoch (t0) 1683.4391 BTJD
Duration 0.0840 d
Depth 8.883 % (88,832 ppm)
SNR 507.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 75 (gate ≥ 3)
Robust depth ratio 1.23 (gate ≥ 0.50)
Robust depth sigma 414.0 (gate ≥ 7.0)
Odd/even z-score 1.61 (consistent)
Secondary eclipse detected, phase -0.49, 7.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 51.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 27.82 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.18 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 237115821
1.72 px away, dTmag +2.50 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

